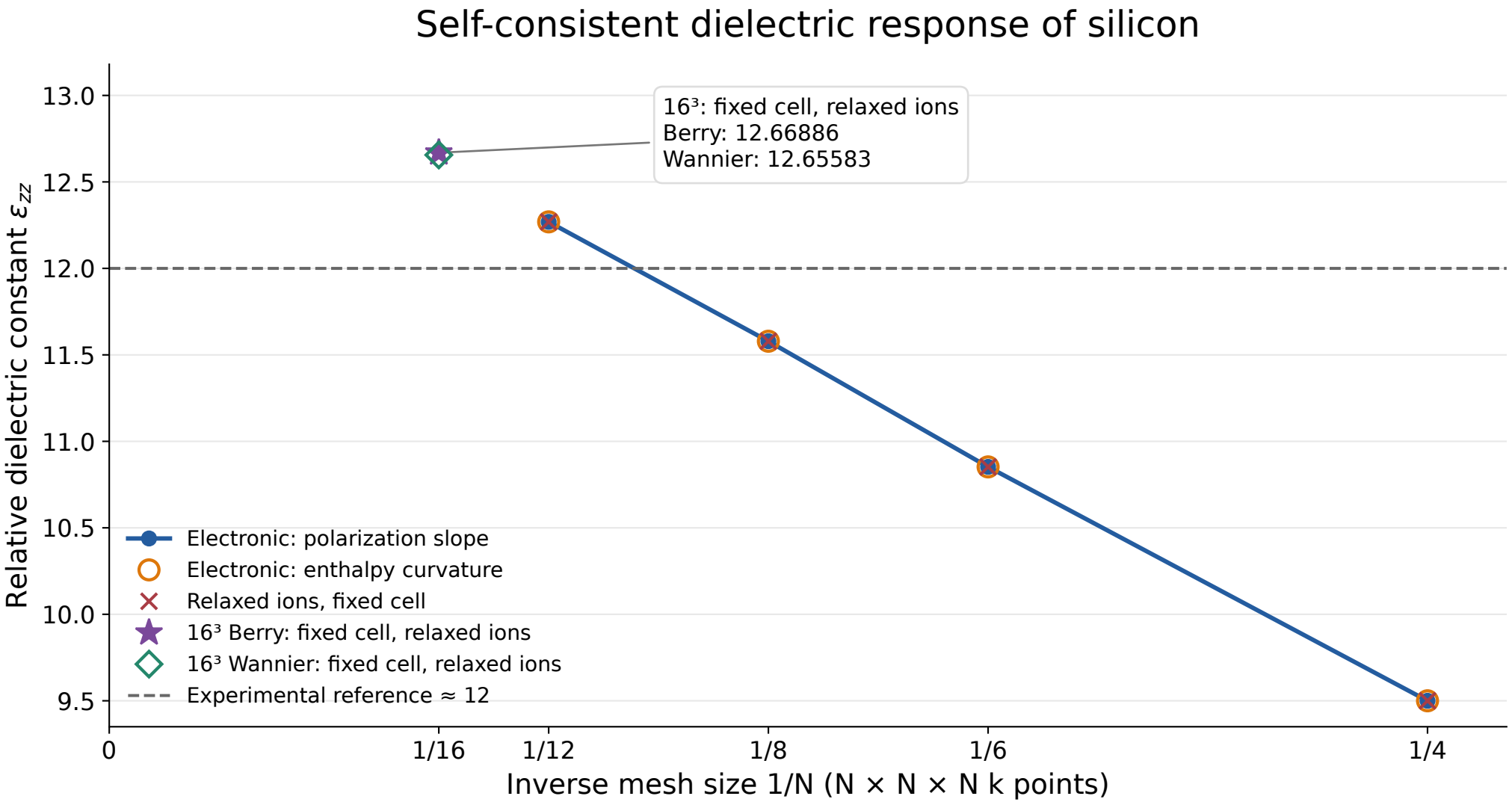
Self-consistent dielectric response of silicon
Relative dielectric constant ϵ_{zz}
13.0
12.5
12.0
11.5
11.0
10.5
10.0
9.5
16³: fixed cell, relaxed ions
Berry: 12.66886
Wannier: 12.65583
Electronic: polarization slope
Electronic: enthalpy curvature
Relaxed ions, fixed cell
16³ Berry: fixed cell, relaxed ions
16³ Wannier: fixed cell, relaxed ions
Experimental reference ≈ 12
0
1/16
1/12
1/8
1/6
1/4
Inverse mesh size 1/N (N × N × N k points)
4³–12³: fixed a = 10.26 bohr; 16³ reference a = 10.350505 bohr.
|Ez| = 0.0001 a.u.; PBE/SG15; cutoff = 30 Hartree. Different geometries: not a mesh-only comparison.